

Project - Security Hardware

Aleksander Klap

08.06.2026

1 Security Hardware

This chapter lists the hardware required to implement the redesigned architecture (Chapter 7).

1.1 Industrial Firewall

Model	Fortinet FortiGate Rugged 70G
Function	Central Layer 3 firewall and inter-VLAN router. All traffic between VLANs passes through it - under default-deny polict. Can perform stateful filtering and packet inspection of industrial protocols (S7comm+, OPC UA, MQTT, HTTP/S). Implements four logical firewalls (Edge, IT-IDMZ, IDMZ-OT Cell, OT Cell-OT Production) as separate rule sets between VLAN interfaces.
Placement	Network core. Has one tagged VLAN interface (.254 gateway) per VLAN. Has single 802.1Q trunk to the managed switch. No VLAN is bridged on the switch, so the firewall is the only inter-VLAN path.

1.2 Managed Switch

Model	Siemens SCALANCE XC-216
Function	VLAN-aware Layer 2 switching with IEEE 802.1Q tagging. Provides per-station access ports (untagged, one VLAN each) and the trunk uplink to the firewall.
Placement	Replace the existing unmanaged SCALANCE XB008 switches at each station. Interconnected by 802.1Q trunks; one trunk reaches the firewall.

1.3 Engineering Station

Model	Siemens SIMATIC IPC627E
Function	Dedicated managed station running TIA Portal for PLC and HMI programming/configuration. Only host permitted S7comm+ to PLCs/HMIs.
Placement	OT Cell zone, isolated in its own VLAN (40). Reaches OT Production only through the firewall on TCP 102.

1.4 IDMZ proxy server

Model	Industrial rack server or SIMATIC IPC. Hardened Linux.
Function	Hosts the IDMZ services — OPC UA proxy, Nginx reverse proxy, MQTT broker, NTP master, log forwarder, backup staging.
Placement	Placed inside IDMZ zone - on VLAN20.

1.5 Syslog & Backup Storage

Model	Standard server, Linux OS.
Function	Central Syslog collector for firewall, switches, proxies and broker; offline configuration backups (after every change) and daily MES database backups over SFTP.
Placement	IT zone, VLAN10